

Day 13 – 3D Modelling & Assets Team

Progress Report

Team 2: 3D Modelling & Assets Team

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Objective

The objective of today's work was to collect and organize 3D assets required for the future development of a Virtual Try-On system. The focus was on understanding the types of assets needed, creating a structured asset repository, exploring human avatar resources, and gaining basic exposure to Blender for 3D asset management.

Asset Repository Creation

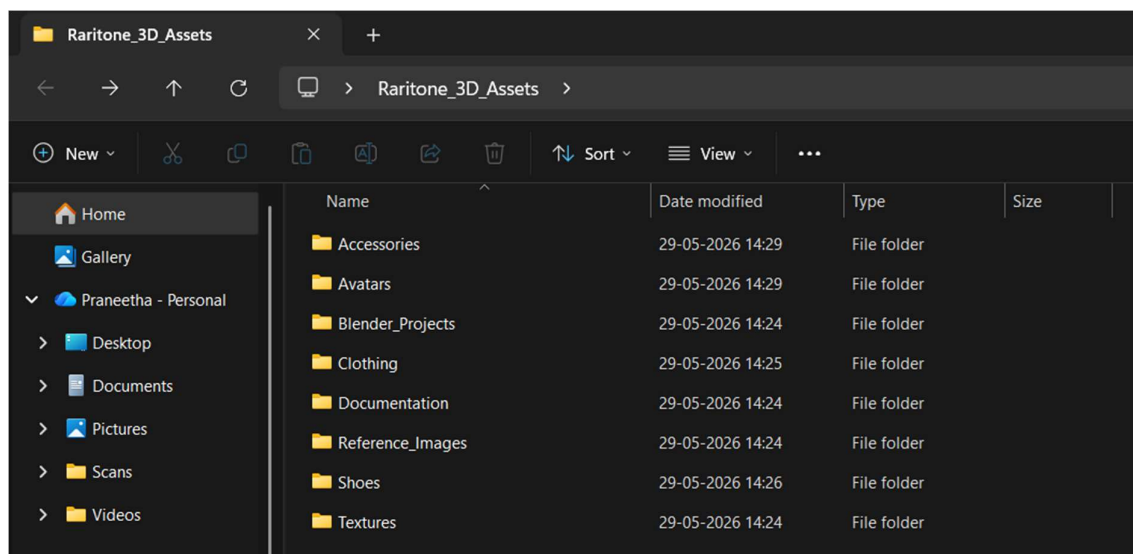
A centralized repository named **Raritone_3D_Assets** was created to organize all resources required for future 3D Virtual Try-On development.

The repository was structured into multiple categories to ensure easy access, scalability, and future integration with Virtual Try-On systems.

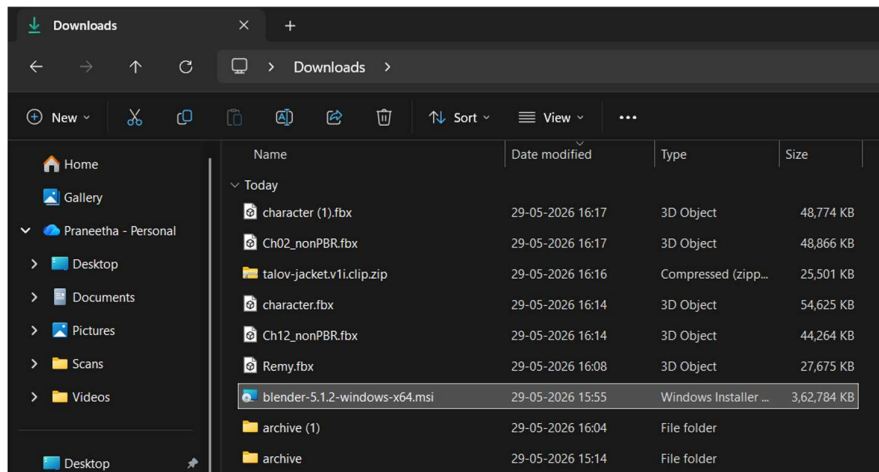
Repository Structure

- Clothing
 - T-Shirts
 - Shirts
 - Jackets
 - Hoodies
 - Dresses

- Shoes
 - Sneakers
 - Formal Shoes
 - Sports Shoes
 - Boots
- Accessories
 - Watches
 - Glasses
 - Bags
 - Caps
 - Jewelry
- Avatars
 - Male
 - Female
- Blender Projects
- Reference Images
- Textures
- Documentation



Download required contents:



Clothing Asset Collection

Fashion datasets were explored and organized to collect clothing assets that can be used for future Virtual Try-On applications.

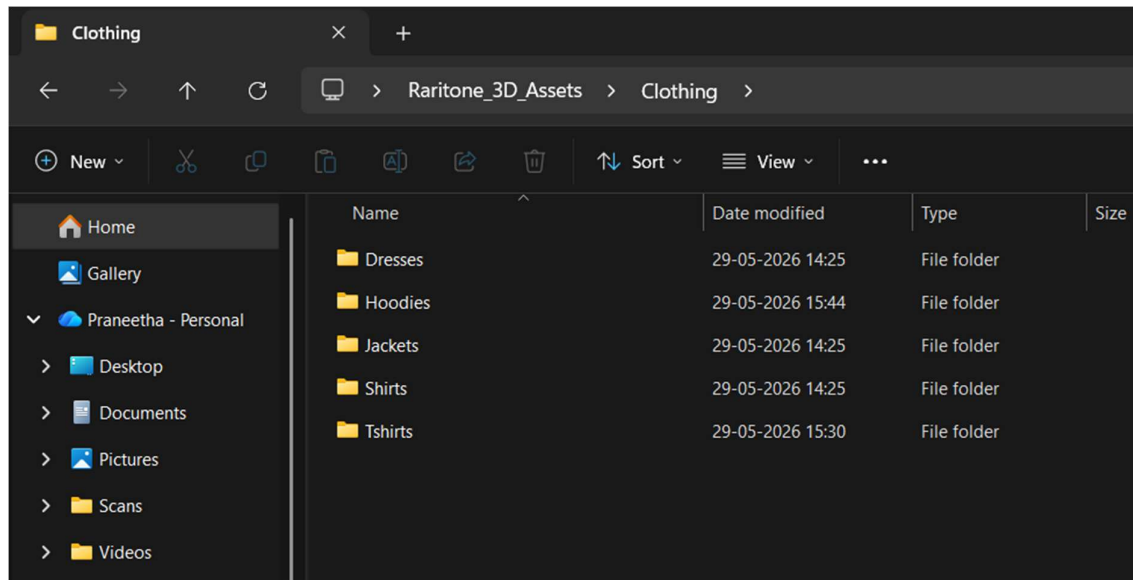
Clothing categories collected include:

- T-Shirts
- Shirts
- Jackets
- Hoodies
- Dresses

The collected assets were categorized and stored in dedicated folders for easier access and future model training or visualization purposes.

Learning Outcome

- Understood different clothing categories used in Virtual Try-On systems.
- Learned the importance of organizing fashion assets for efficient development.
- Explored fashion datasets commonly used in AI and computer vision projects.



Footwear Asset Collection

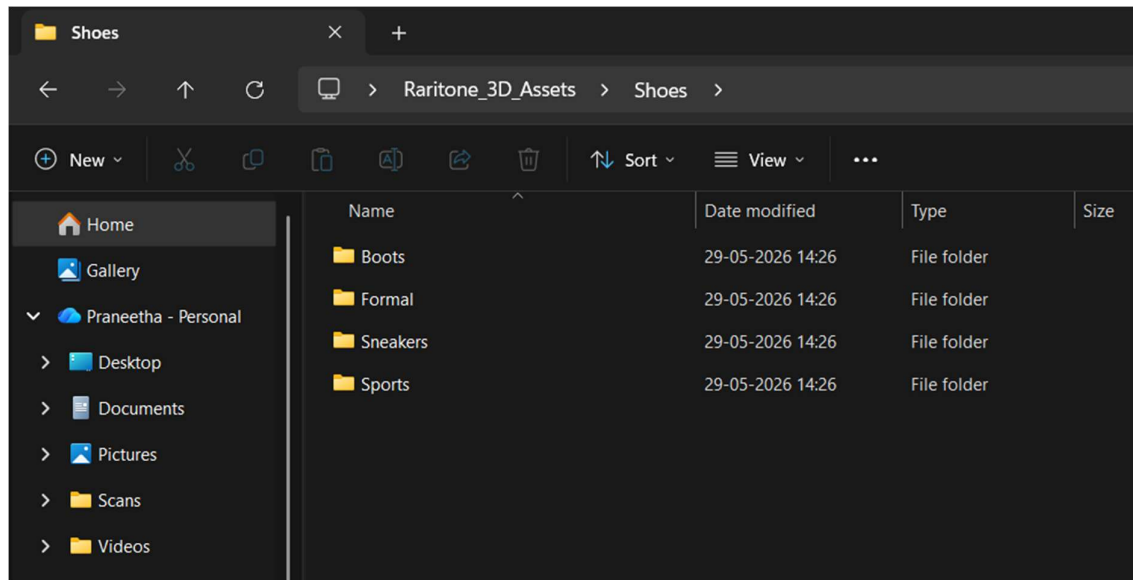
Footwear assets were collected and categorized into:

- Sneakers
- Formal Shoes
- Sports Shoes
- Boots

These assets will be useful for future extensions of the Virtual Try-On platform where footwear visualization and recommendation features may be integrated.

Learning Outcome

- Learned how footwear assets are organized for digital fashion applications.
- Understood the role of footwear visualization in complete virtual styling systems.



Accessories Collection

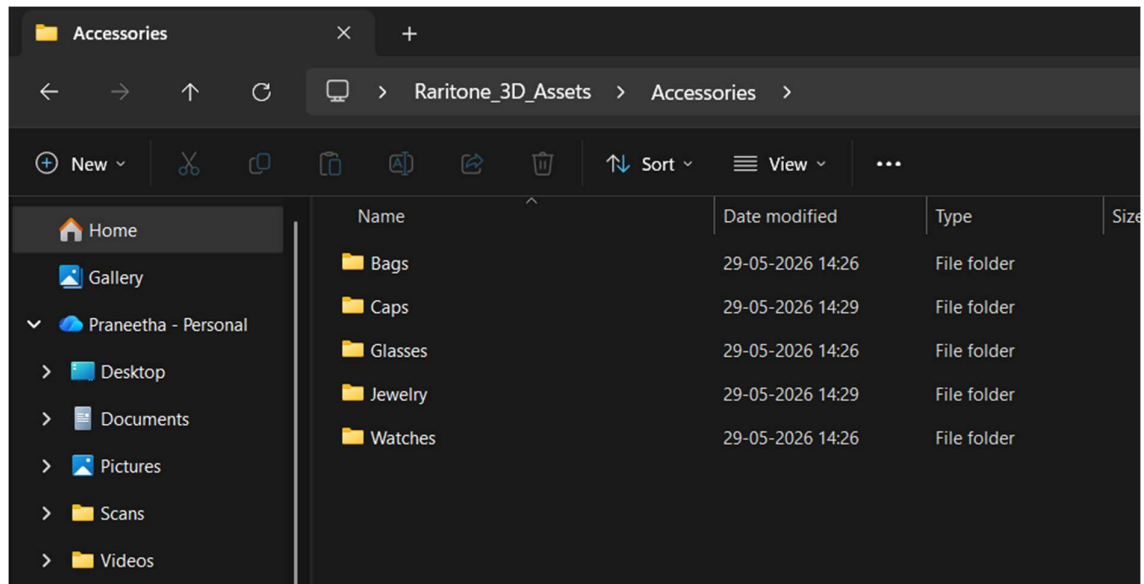
Accessories were organized into multiple categories:

- Watches
- Glasses
- Bags
- Caps
- Jewelry

The purpose of collecting these assets is to support future personalization and styling features within the Virtual Try-On platform.

Learning Outcome

- Explored different fashion accessory categories.
- Understood how accessories enhance user personalization and virtual styling experiences.



Human Avatar Collection

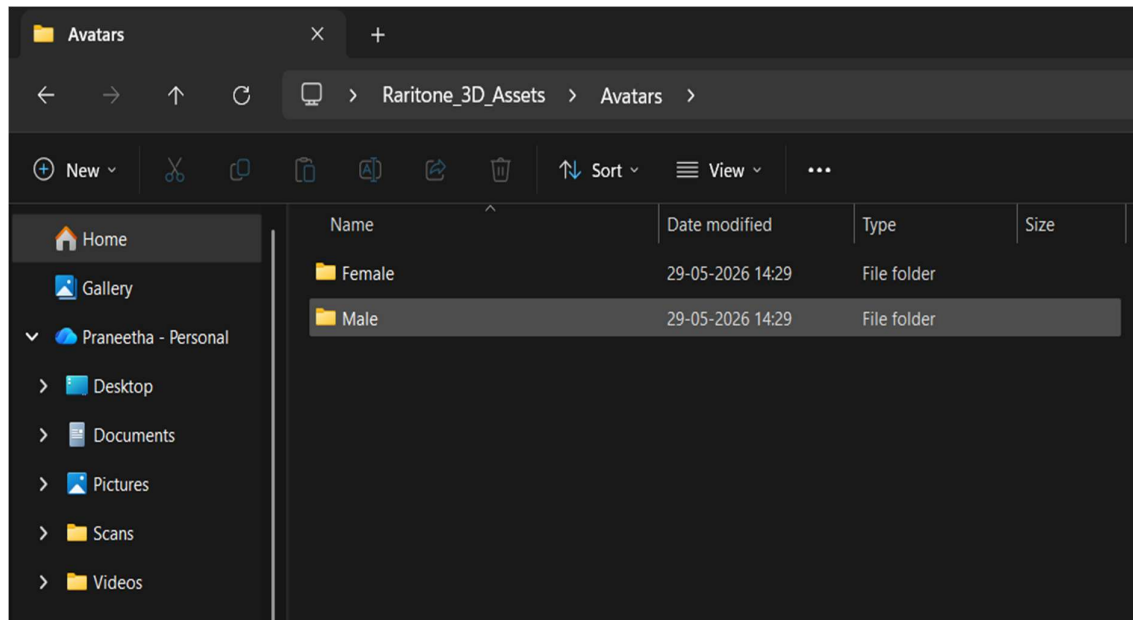
Human avatar resources were explored and organized into:

- Male Avatars
- Female Avatars

These avatar assets serve as the digital representation of users in Virtual Try-On environments and can later be integrated into Blender, Unity, or AR/VR systems.

Learning Outcome

- Learned the importance of human avatars in 3D Virtual Try-On systems.
- Explored avatar formats and their role in virtual clothing visualization.



Blender Exploration

Blender was explored as a 3D modeling and visualization tool.

Activities performed:

- Opened and explored the Blender interface.
- Understood the basic workspace layout.
- Learned about importing 3D models.
- Explored how avatars and clothing assets can be visualized within a 3D environment.

Although no advanced modeling was performed, this exploration provided an understanding of how Blender can be used in future stages of Virtual Try-On development.

Learning Outcome

- Familiarized with Blender's interface.
- Understood the workflow for importing and viewing 3D assets.
- Learned how Blender can support avatar visualization and asset management.

Figure 1: Blender default workspace

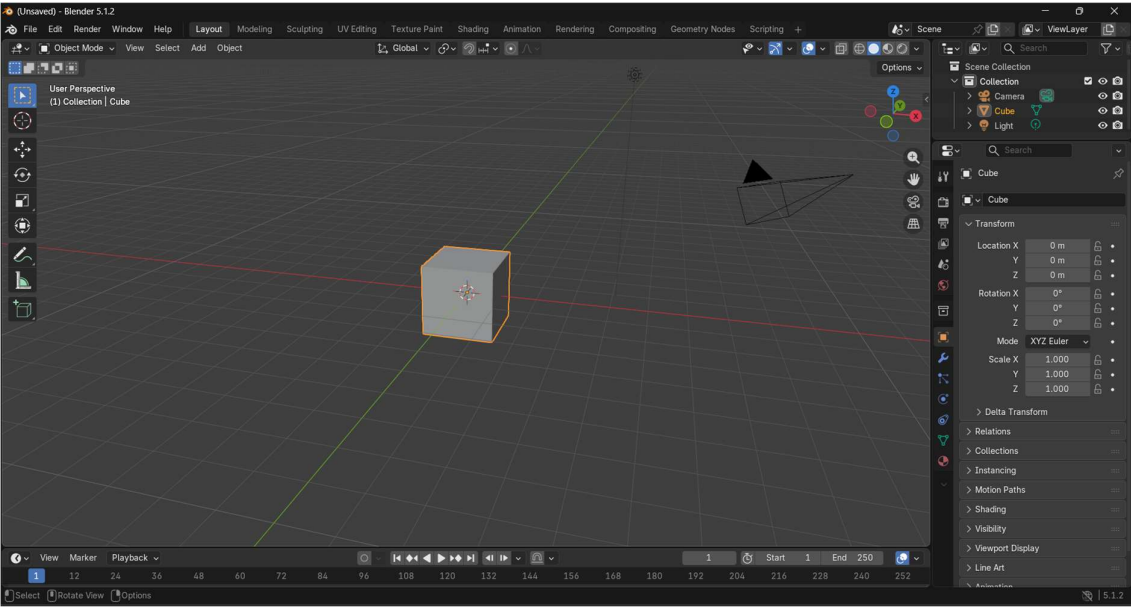


Figure 2: Avatar view

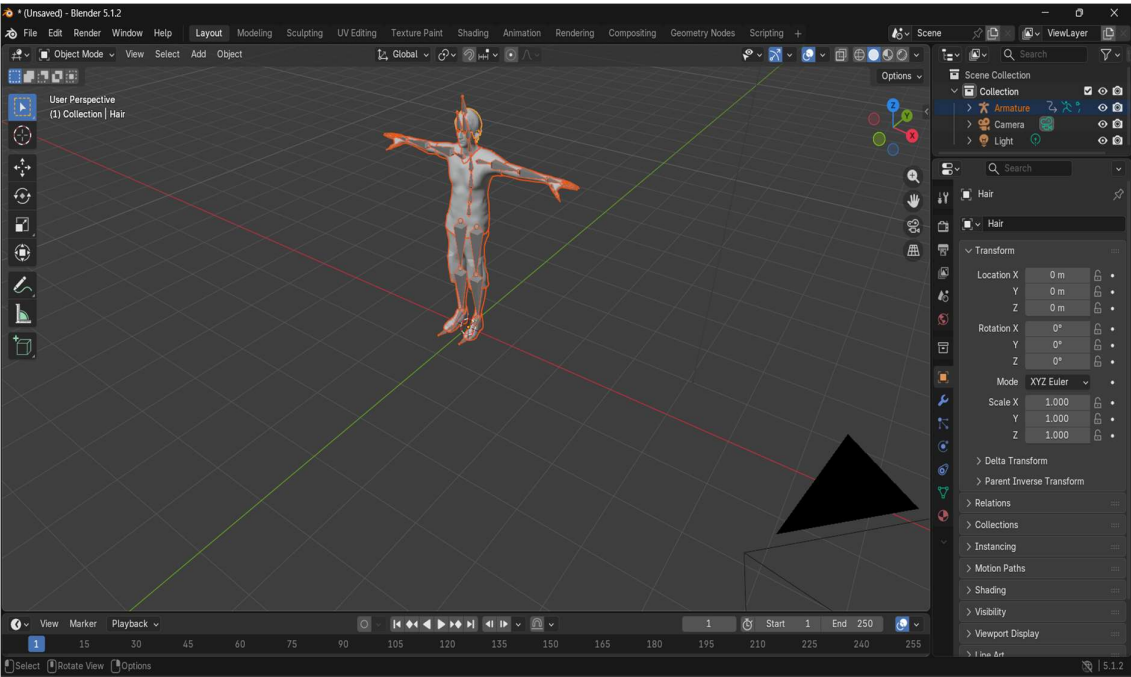


Figure 3: Camera view:

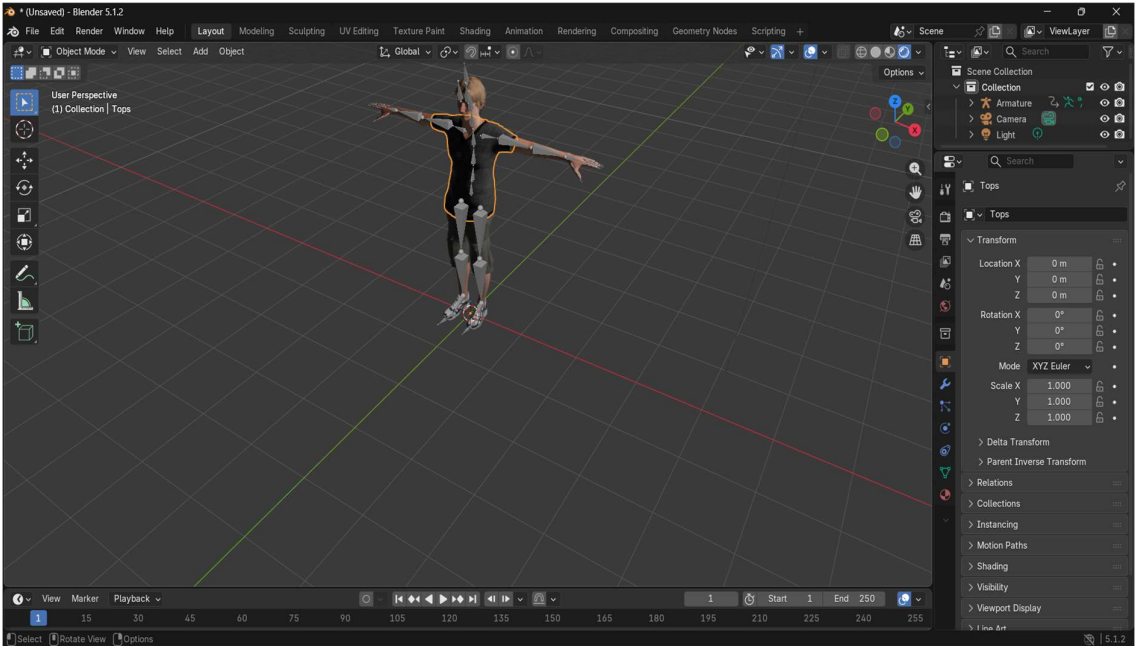
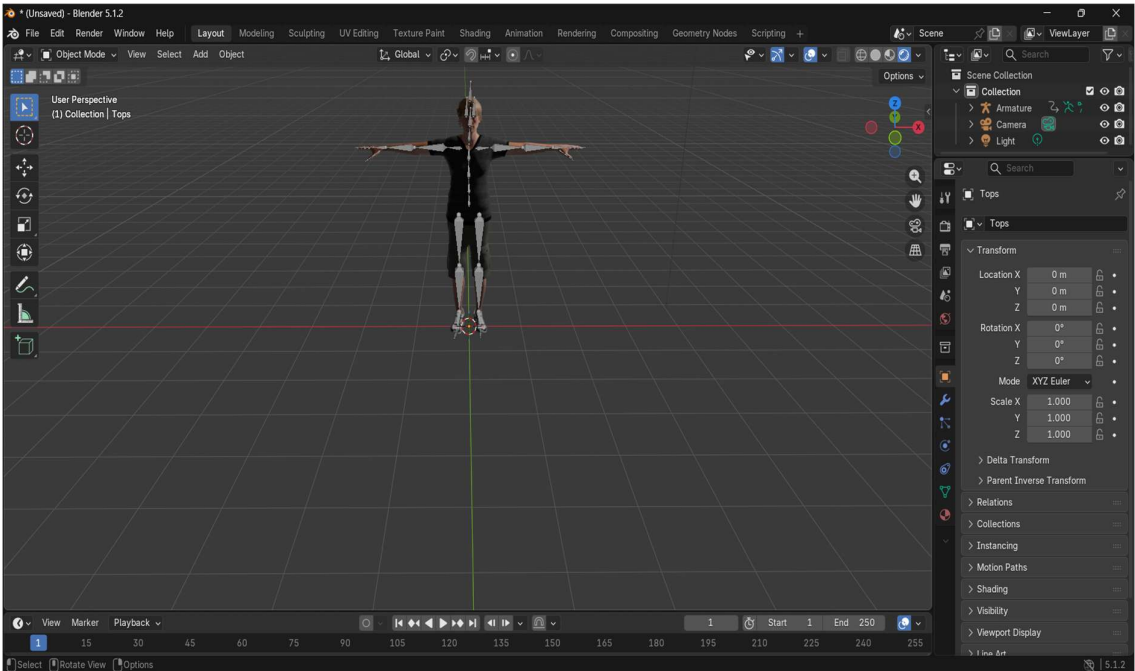


Figure 4: Front View



Key Learnings

- Importance of maintaining a structured asset repository.
- Different categories of assets required for Virtual Try-On systems.
- Basics of human avatar management.
- Introduction to Blender for 3D asset visualization.
- Understanding the relationship between clothing assets, avatars, and Virtual Try-On applications.

Future Scope

The collected assets and repository structure will support future development activities such as:

- 3D avatar customization
- Cloth simulation
- Body mesh fitting
- Real-time Virtual Try-On systems
- AR/VR-based fashion applications
- Unity and Blender integration

Conclusion

Today's work focused on building the foundation required for future 3D Virtual Try-On development. A structured asset repository was created, fashion assets were collected and categorized, avatar resources were explored, and Blender was introduced as a tool for 3D visualization. The activities completed today provide the necessary groundwork for implementing more advanced Virtual Try-On and AR/VR features in future stages of the project.